

Although some researchers felt that Thurston's experiment was too hypothetical, it was still considered important. In the 1940s, two researchers named Stephen W. Rousseas and Albert G. Hart performed some experiments on indifference curves designed to follow up on Thurstone's experiment and to respond to some of the experiment's critics. They constructed what they viewed as a more concrete and realistic choice situation by having subjects select among possible breakfast menus, with each potential breakfast consisting of a specified number of eggs and a specified quantity of bacon strips. They required that "each individual was obliged to eat all of what he chose; i.e., he could not save any part of the offerings for a future time."⁵

In this experiment, individual subjects made only a single choice (repeated subsequently a month later); and, in addition to selecting among available combinations, each was asked to state an ideal combination of bacon and eggs. While this experiment did not ask its subjects to make too many choices of the same type (i.e., different combinations of two goods), thereby averting a common criticism of Thurstone, it left Rousseas and Hart with the problem of trying to aggregate individual choice data collected from multiple individuals. They attempted to ascertain whether choices made by separate individuals stating similar "ideal" breakfast combinations could be pieced together to form consistent indifference curves. This last step presented complications, but overall the project was considered a success and led to further experiments in the same vein.

Also inspired by Thurstone, Frederick Mosteller and Phillip Noguee sought in 1951 to test expected utility theory by experimentally constructing utility curves.⁶ Mosteller and Noguee tested subjects' willingness to accept lotteries with given stakes at varying payoff probabilities. They concluded, in general, that it was possible to construct subjects' utility functions experimentally and that the predictions derived from these utility functions were "not so good as might be hoped, but their general direction [was] correct." This is a conclusion that many experimental economists would still affirm, with differing degrees of emphasis.

As these types of experiments continued, various violations of expected utility were beginning to be observed. Perhaps the most famous of violations of expected utility was exposed by another Nobel Prize winner in Economic Sciences in 1988.

Maurice Allais (Figure 2.3), Memorial Prize in Economic Sciences winner (1988), made distinguished, pioneering, and highly original contributions in many areas of economic research. Outside of a rather small circle of economists, he is perhaps best known for his studies of risk theory and the so-called Allais paradox. He showed that the theory of maximization of expected utility, which had been accepted for many decades, did not apply